



Funnel Analysis and User Behavior Insights

A data analytics project examining user conversion performance across a simulated 30-day e-commerce journey.

Presented By

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Subject

Python(Data Analytics)

Focus

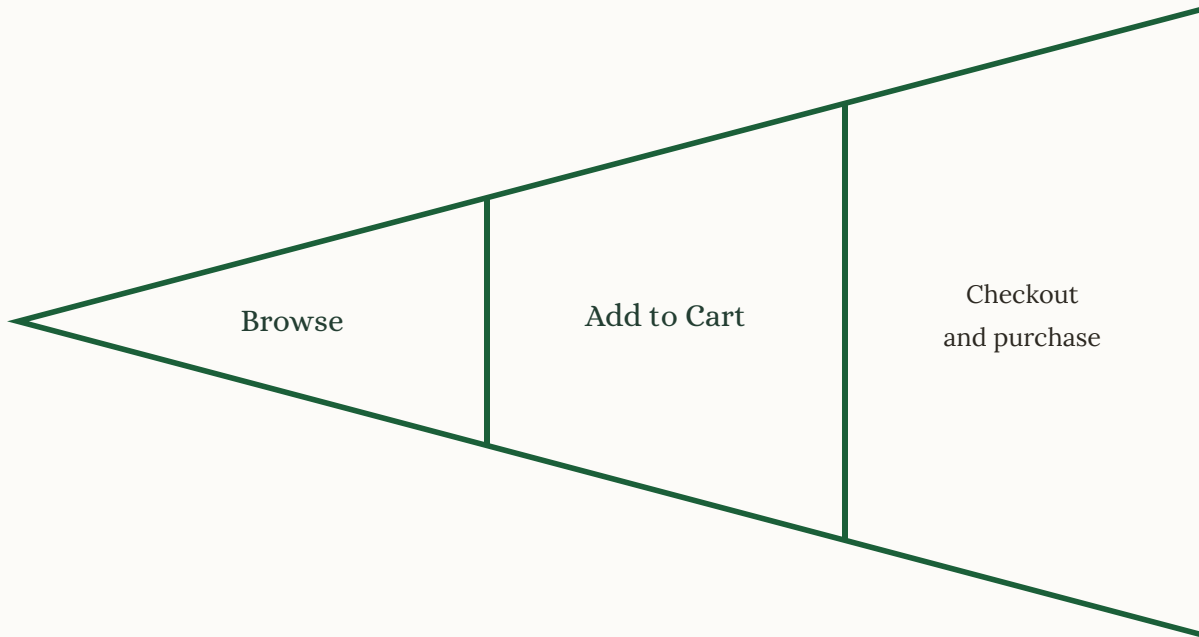
Conversion Optimization

Submitted to :

Dr. Sunesh Malik

What is Funnel Analysis?

Funnel analysis tracks how users progress through the sequential stages of a customer journey — from first interaction to final conversion. At every stage, some users drop off, and understanding **why** is the foundation of conversion optimization.



Businesses lose revenue at every stage. Identifying and addressing these drop-off points is critical for digital product growth and e-commerce profitability.



Project Objectives :-

1

Analyze the User Journey

Trace the complete path across all funnel stages from browse to purchase.

2

Identify Drop-Off Points

Pinpoint exactly where and why users abandon the funnel.

3

Behavioral Segmentation

Understand patterns by device, marketing channel, and geographic region.

4

Data-Driven Recommendations

Provide actionable insights to improve conversion rate and revenue.

Dataset Overview

A **synthetic dataset** of ~10,000 user sessions was generated using the Python `Faker` library, simulating 30 days of e-commerce activity across devices, regions, and marketing channels.



Time Period: 30 days | Total Records: ~10,000 sessions

Data generation and preprocessing were performed in Visual Studio Code.

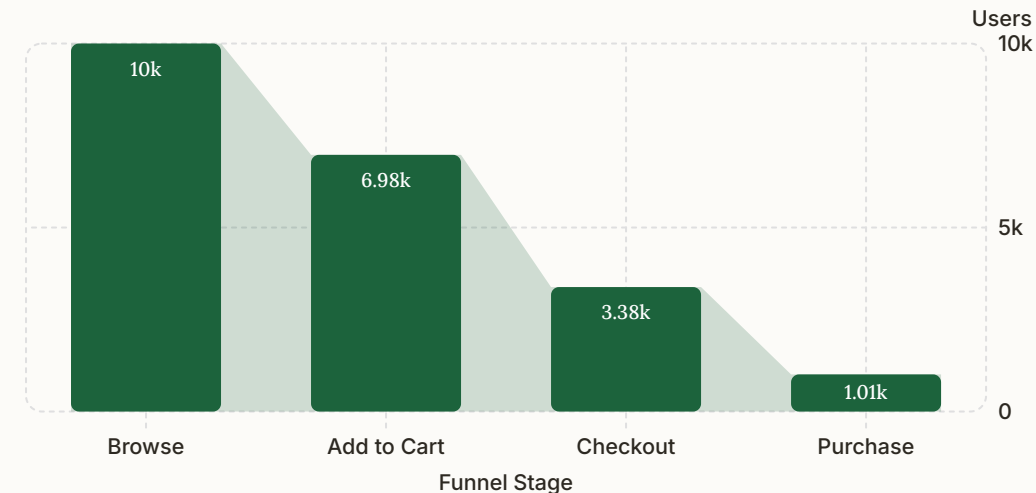
The dataset simulates real-world user behavior in an e-commerce funnel, including events like Browse, Add to Cart, Checkout, and Purchase.

Column	Description
User_ID / Session_ID	Unique user and session identifiers
Event	Funnel stage: Browse, Add to Cart, Checkout, Purchase
Device	Mobile, Desktop, Tablet
Region	North, South, East, West
Channel	Google Ads, Organic, Email, Social Media
Product_Category	Electronics, Fashion, Home, Beauty, Sports
Revenue	Purchase amount (Purchase events only)
Bounce_Flag	Yes / No — did user leave immediately?

Funnel Analysis(Total Conversion Rate):-

10.09%

Total conversion rate



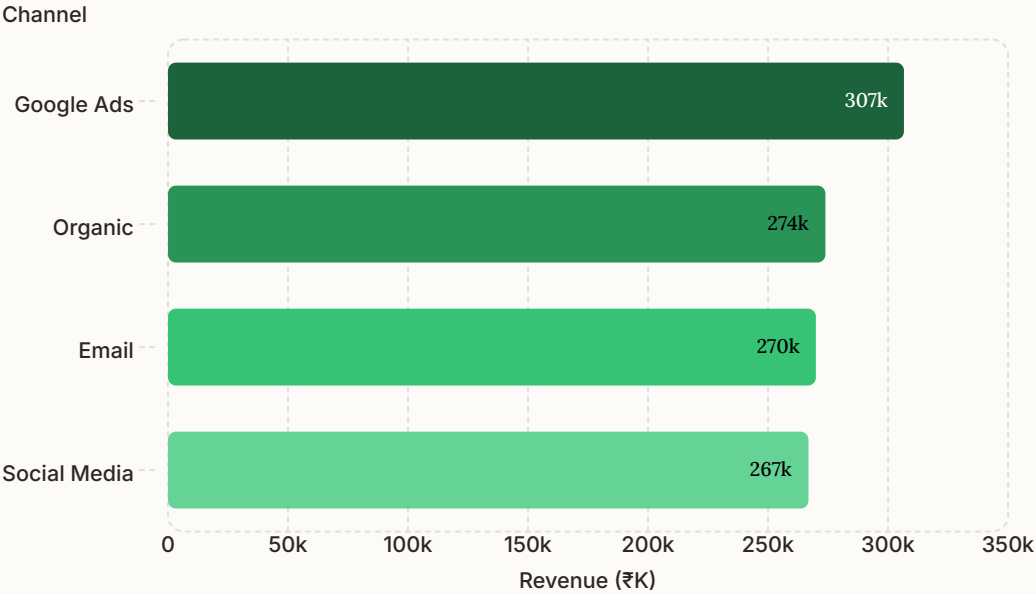
Stage	Users	Conversion	Drop-off
Browse	10,000	100%	—
Add to Cart	6,975	69.75%	30.25%
Checkout	3,382	33.82%	35.93%
Purchase	1,009	10.09%	23.73%



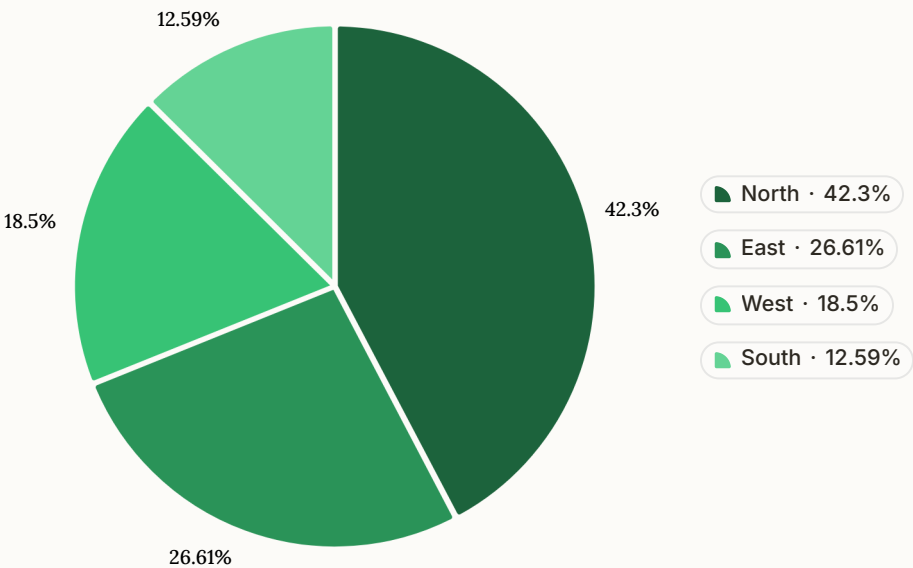
The biggest drop-off occurs between **Browse** → **Add to Cart** (30%) and again at **Checkout** → **Purchase**. Overall conversion: ~10%.

Revenue & Channel Performance

Marketing Channel Revenue



Regional Revenue Distribution



The **North region dominates** with 42.3% of total revenue. Google Ads is the top-performing acquisition channel.

Time-Based Revenue Trends

₹1.17M

Total Revenue

Aggregated across all 1,009 purchase sessions

~₹1,100

Avg Order Value

Revenue per purchase ranging ₹200–₹2,000

51K

Peak Day Revenue

Day 23 recorded the highest single-day revenue

2K

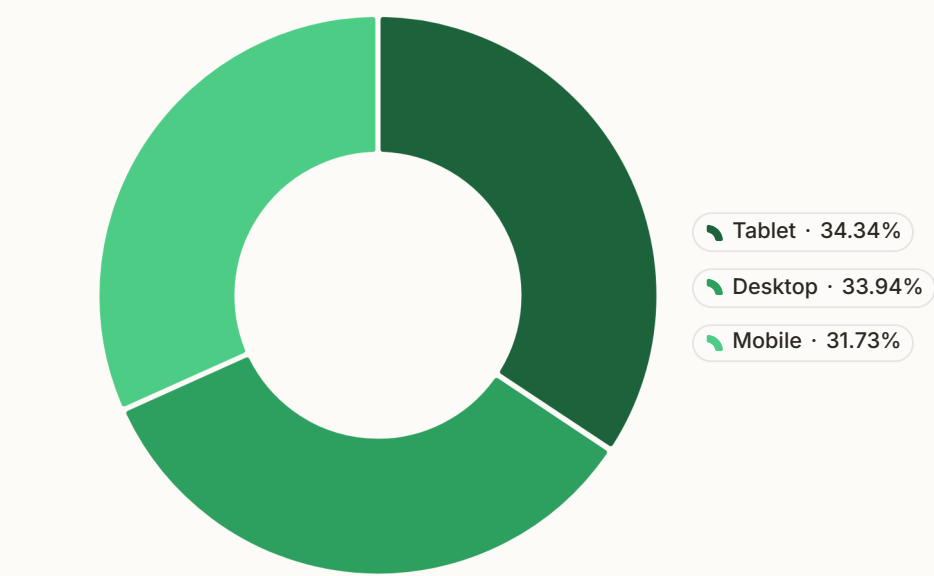
Lowest Day Revenue

Day 13 — a significant anomaly in the 30-day window

Revenue is **volatile across the 30-day period**, with wide daily fluctuations likely tied to campaign schedules and weekend traffic patterns. Hourly analysis further pinpoints optimal windows for promotions.

Device & Product Category Analysis

User Distribution by Device



- 1 Electronics
- 2 Fashion
- 3 Home
- 4 Beauty
- 5 Sports

Key Device Insight

All three devices show a **~90% drop-off rate** across the funnel — nearly identical across Mobile, Desktop, and Tablet.

i This uniformity reveals the drop-off is a **funnel-design issue**, not a device-specific one. The problem lies in the experience itself.

Product Categories Analyzed

Key Findings & Recommendations

Key Findings

High Checkout Drop-Off

77.21% of users who reach checkout do not complete a purchase.

Google Ads Leads

Highest revenue contribution among all acquisition channels.

North Region Dominates

42.3% of total revenue — a prime target for premium campaigns.

Device-Agnostic Drop-Off

90% drop-off is consistent across all devices — a systemic funnel issue.

Recommendations

→ Simplify Checkout

Reduce friction with fewer steps, guest checkout, and saved payment options.

→ Double Down on Google Ads

Reallocate budget toward the highest-converting acquisition channel.

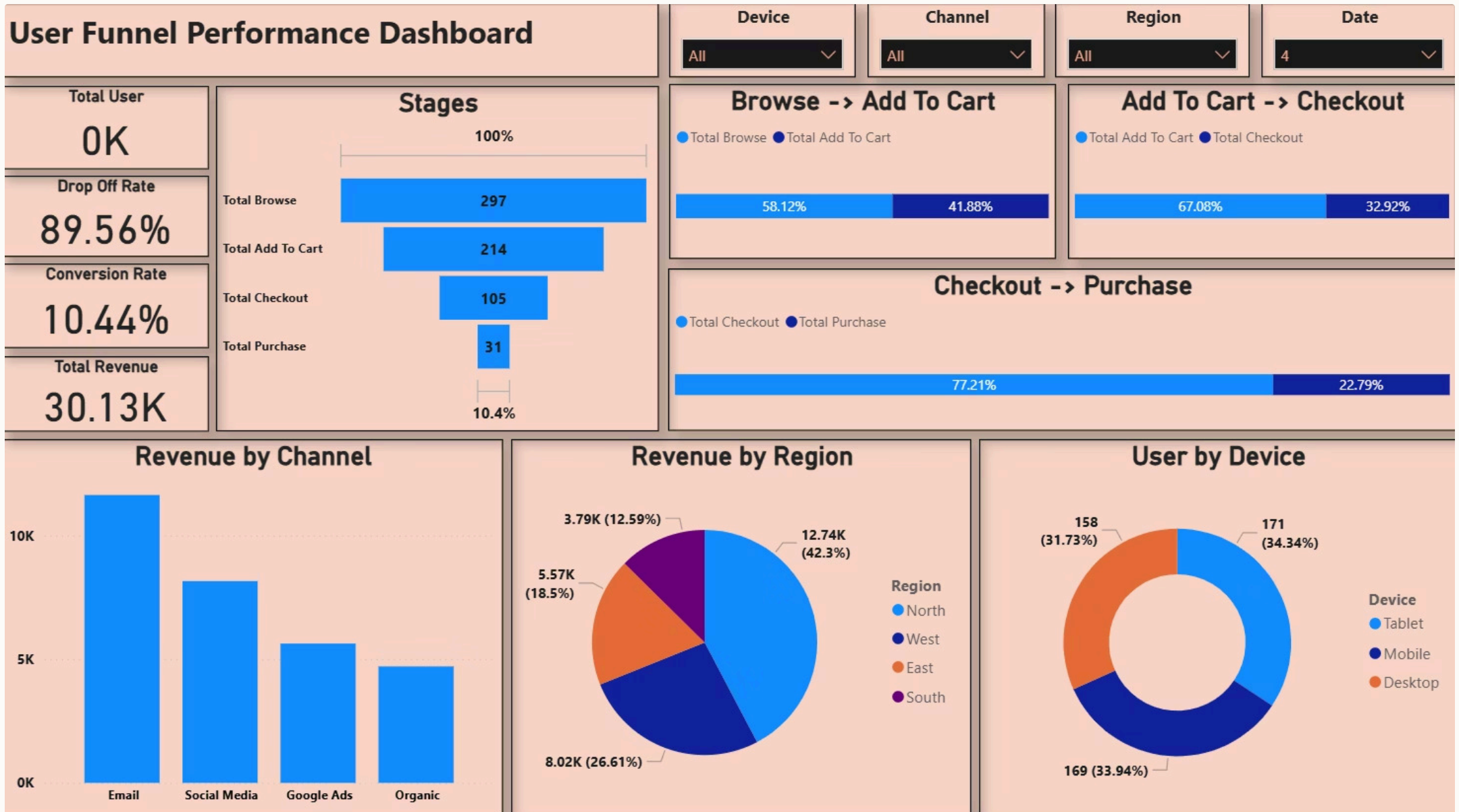
→ Retargeting Campaigns

Launch cart abandonment flows via email and social to recover lost revenue.

→ Optimize Campaign Timing

Align promotions with peak hours and days identified in time-based analysis.

User Funnel Performance dashboard



Funnel Drop Off Dashboard

Funnel Drop-Off Diagnostics

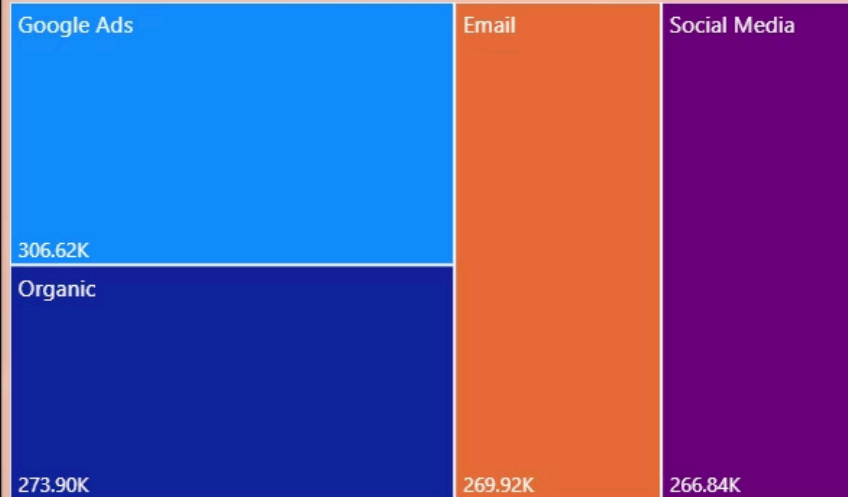
% Browse
100.00%

% Add to cart
69.75%

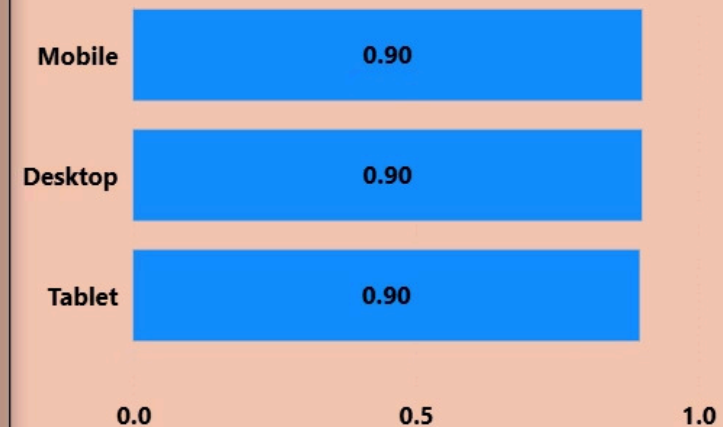
% Checkout
33.82%

% Purchase
10.09%

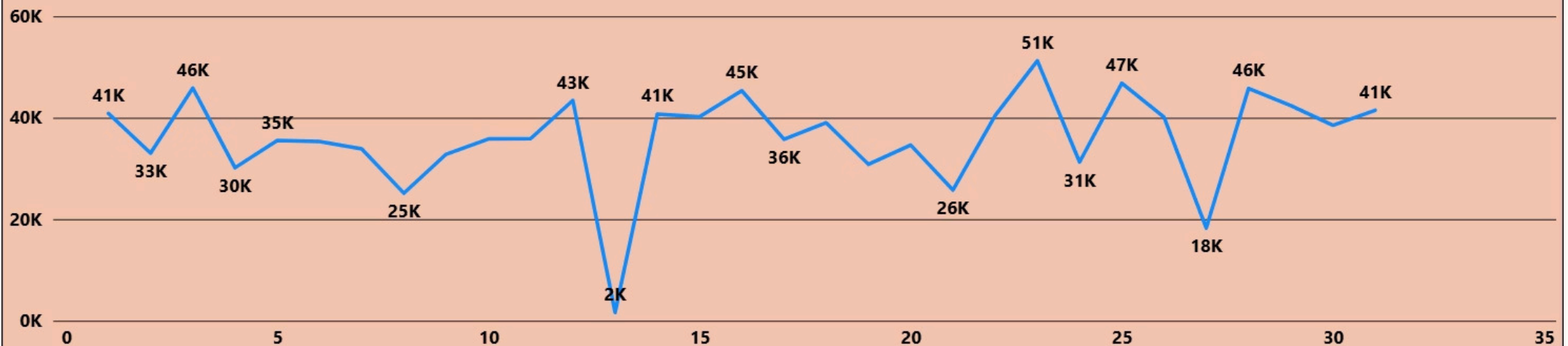
Revenue by Channel



Drop Off Rate by Device



Sum of Revenue by Day



Conclusion

This project delivered a full-spectrum analysis of user behavior across a simulated 30-day e-commerce funnel, uncovering a **~10% overall conversion rate** and actionable patterns across channels, regions, and devices.

10,000+ Sessions

Analyzed across a 30-day simulated period

Power BI Dashboards

Interactive stakeholder-ready visuals built end-to-end

Python · Pandas · Faker

Full analytics pipeline from data generation to insight

Thank you — Shubh Shekhar Jha | Enrollment No: 02096303123